abilities, such as an operator who can "hear" when a pump is developing cavitation (the implosion of vapor bubbles sounds like gravel moving through the pump), or a facilities manager who knows that the east-wing HVAC system always struggles during afternoon sun exposure.

The challenge becomes even more complex in complex industrial settings. A chemical processing plant might have decades of operational knowledge spread across retiring engineers, tribal knowledge embedded in informal procedures, and hard-won insights from equipment failures that happened years ago. Capturing this knowledge enables a digital twin to effectively inform decision making and ultimately autonomous operation in the way a skilled operator can.

NOTE

The risk posed by the wave of retirement of skilled workers in manufacturing and associated loss of knowledge and expertise has already seen companies such as Boeing having to rehire retired engineers to ensure timely delivery from its 737 production line.

As an aging workforce moves to retirement across the industrialized world, the ability to capture, store, and use decades of accumulated knowledge and wisdom offers a huge potential benefit to many organizations.

5.2 Understanding context in digital twins

In the previous section, we explored how the knowledge engineering challenge requires us to capture and systematically represent the contextual understanding that human experts apply naturally. But what exactly do we mean by "context" in digital twin systems, and why is it so critical for creating intelligent, actionable insights?

Consider the example in figure 5.3 where a reading of 85° is received from a temperature sensor. In the case of the reading being related to an industrial furnace, this can be interpreted as being completely normal, based on a number of related pieces of data, most importantly that this is the optimal